NOTES

Paleomagnetism of Igneous Rock Units from the Coast of Labrador

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Paleomagnetic results are reported for the first time from the coast of Labrador. The rocks are from (*1*) 20 sites of gently folded basalts of the Mugford Series, dated 948 ± 90 m.y., near Cape Mugford, and (*2*) 12 gabbroic dikes in a swarm dated 2080 ± 42 m.y., near Indian Harbour. Both dates are from preliminary, published K–Ar determinations. After partial alternating-field (AF) or thermal demagnetization, nearly all samples from both rock units revealed a stable thermoremanence that is most probably of primary origin. For the Mugford basalts, a mean remanence direction relative to bedding planes was calculated, after AF treatment, from 16 stably magnetized sites (≧16 flows). The corresponding pole is 49° N, 143° W, with dp = 9°, dm = 11°, which is northeastward of most North American poles from rocks dated 1200-1000 m.y. The pole calculated from the 12 Indian Harbour dikes, after AF treatment, is 6° S, 117° W, with dp = 6°, dm = 12°.

C'est la première fois que des résultats paléomagnétiques relatifs à la côte du Labrador sont rapportés dans la littérature. Les roches viennent de (1) 20 gisements de basaltes légèrement plissés de la Série Mugford, près du Cap Mugford, et datés 948 ± 90 millions d'années, et (2) de 12 filons gabbroiques d'un essaim près de "Indian Harbour" et datant de 2080 ± 42 millions d'années. Les deux âges ci-haut mentionnés furent déterminés par la méthode K–Ar et ont déjà fait l'objet de publications préliminaires par des autres auteurs. Après une démagnétisation partielle par champ alternatif ("alternating-field ou AF") ou thermique, presque tous les échantillons des deux séries ont conservé une thermorémanence fort probablement primaire. Dans le cas des basaltes de Mugford, la direction rémanente moyenne relative au litage fut calculée, après un traitement AF, à partir de 16 gisements ayant une thermorémanence stable (représentant ≧16 coulées). Le pôle correspondant a une attitude de 49° N, 143° W avec dp = 9°, dm = 11°. Cette valeur est au nord-est de la majorité des pôles de roches datant de 1200–1000 millions d'années en Amérique du Nord. Le pôle calculé à partir de 12 filone, après un traitement AF, a une attitude 6° S, 117° W avec dp = 6° et dm = 12°.

Introduction

Many paleomagnetic results are now available from radiometrically dated Precambrian rocks in North America. From his pioneering studies of Keweenawan rocks bordering Lake Superior, DuBois (1962) obtained poles distributed about a long arc in the present Pacific, which he interpreted as polar displacement relative to North America. Although more recent results from the Lake Superior area (summary in Beck 1970) and the western part of the Canadian Shield (summary in Fahrig and Jones 1969) are broadly concordant with DuBois' data, their grouping is such that neither a fine structure nor even a trend of apparent polar wandering can yet be established for the 1200–1000 m.y. age span. These uncertainties call for additional results from rocks that were magnetized about one billion years ago. A virtually untapped source of new paleomagnetic data is the Atlantic margin of the Canadian Shield, and in this note we report the first results from the gently folded lavas of the Mugford Series near Cape Mugford, Newfoundland. From a single whole-rock K–Ar determination on Mugford basalt, Wanless *et al.* (1966) obtained 948 ± 90 m.y. and interpreted this as the age of eruption of the Mugford volcanics. While this date corresponds to the time of the Grenville orogeny, the Mugford area lies 500 km north of the Grenville Front and hence was not involved in the Grenville metamorphism.

The second new result we are reporting here is from a dike swarm near Indian Harbour, north of Hamilton Inlet, Newfoundland. From one of these dikes, composed of apparently fresh

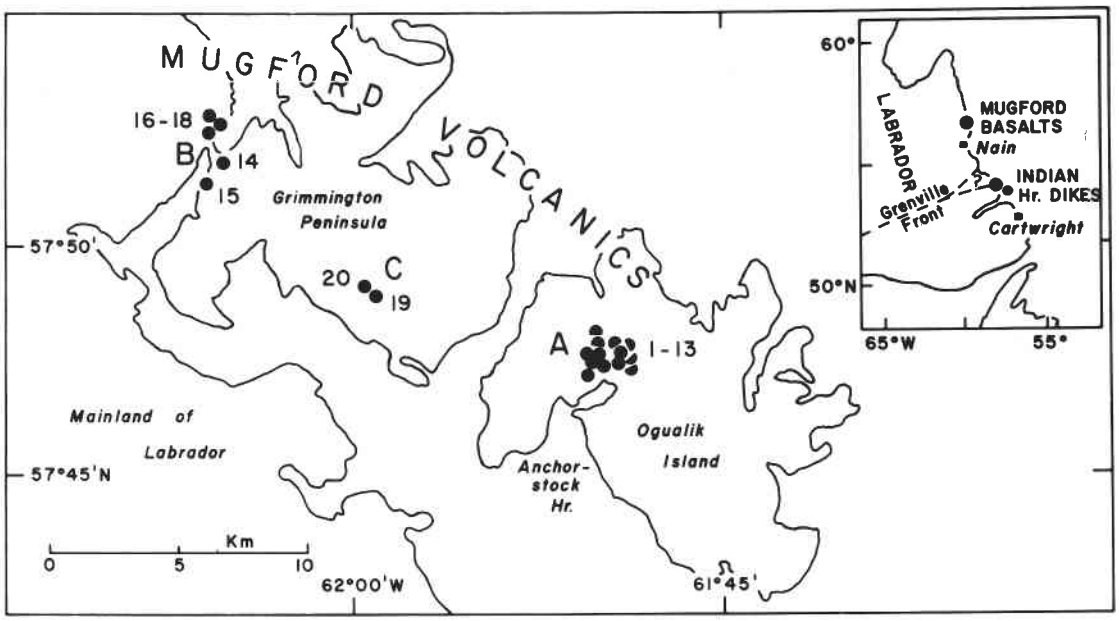


FIG. 1. Paleomagnetic sampling localities on the Labrador coast. The Mugford region is shown in detail; A, B, C denoting sampling areas. Sites are numbered 1-20, each site representing at least one flow. The inset shows the Mugford and Indian Harbour localities.

olivine gabbro, Grasty *et al.* (1969) obtained a whole-rock K–Ar age of 2080 ± 42 m.y. This date suggests that the Indian Harbour dikes could be compared paleomagnetically with the Nipissing diabase in Ontario (2140 ± 60 m.y.), for which a pole position in the eastern Pacific is now well established (Symons 1970). However, Grasty *et al* (1969) believed that their Indian Harbour date might be anomalously high, possibly through argon addition; hence caution is needed in using the magnetic results for long-distance correlation.

Geology and Sampling

The Mugford Volcanic Series forms the higher parts of the Kaumajet Mountains, covering an area of about 400 km² centered on 57.8° N, 62.0° W (Fig. 1). Kranck (1939) and Douglas (1953) have described the geology. The basement is Archean gneiss. A volcanic sequence several hundred meters thick consists mainly of basalt surmounted by andesite and is cut by some vertical basic dikes. There are many more flows than the twenty-five counted by Douglas above Anchorstock Harbour. Apart from faults and a gentle, overall northwesterly tilt, the rocks show little evidence of post-volcanic disturbance. At several places, the Mugford volcanics occur close to the mainly sedimentary Ramah Series, for which Wanless *et al.* (1970) reported a K–Ar date on andesite of 1180 ± 60 m.y.

In the summer of 1968, one of us (E.R.D.) collected oriented block samples of Mugford basalt from 20 sites in three areas A, B, C (Fig. 1). Each site represents at least one flow, without stratigraphic overlap between sites in any area and, most probably, without overlap between different areas. Thus we believe that at least 20 flows were sampled, spanning about 400 m vertically. Criteria used in distinguishing flows include the presence of weathered interfaces and amygdular lava tops. Although the stable results for all three Mugford areas were similar, we shall exclude area C in computing the mean direction (Table 1), since several tens of meters of slate occur between the two flows sampled in that area. In areas A and B, sampling was from continuous lava sequences and, using four sets of observations in these areas, we inferred a mean strike and dip of 250° ± 26, 16° NW ± 2, for the basalts.

The samples were oriented with a Brunton compass or, on bright days, a solar compass. To detect possible Brunton deflection errors due

TABLE 1. Remanence directions and poles for the rock units

Rock unit	Remanence[1]					Paleomagnetic pole			
	D	I	N	k	α_{95}	Lat.	Long.	dp[2]	dm[2]
Mugford basalts (relative to horizontal plane)	263	+71	16	41	6				
Mugford basalts (relative to bedding plane)[3]	297	+63	16	41	6	49° N	143° W	9°	11°
Indian Harbour swarm	242	+22	12	15	12	6° S	117° W	6°	12°
Single dike[4], 958 m.y.[5]	82	−47	4	111	9				

NOTES: 1. After demagnetization to 150 Oe (Mugford) or 300 Oe (Indian Harbour). D = declination in degrees east of north; I = inclination in degrees, always positive (downward); N = number of sites (Mugford), dikes (Indian Harbour), or samples (single dike); k = best estimate of Fisher's precision parameter k; α_{95} = radius of 95% confidence circle.
2. dp, dm = semiaxes of the 95% confidence oval.
3. After correction for the geological dip (see text).
4. NRM data.
5. Dated by Grasty *et al.* (1969).

to the rocks, 15 samples from several sites were oriented with both compasses. The 15 pairs of readings showed excellent agreement, with standard deviation 0.8°. In the laboratory, two or more cylindical specimens 2.2 cm in diameter and 2 cm long were cut from each of 56 blocks. The average remanence of two specimens was taken to represent the sample.

Professor J. C. Rucklidge of the University of Toronto has kindly supplied a set of oriented core samples he collected in 1967 on the Labrador coast, including one sample each from twelve east–west striking dikes of the Indian Harbour swarm located at 54.4° N, 57.1°W (Fig. 1 inset, large dot) and four samples from a meta-lamprophyre dike (small dot), for which we also report a result here. This dike was K–Ar dated 958 ± 19 m.y. by Grasty *et al.* (1969).

Results of Measurements

The natural remanent magnetization (NRM) of all Mugford samples was measured. The intensities (J) of the specimens are weak and are distributed between 2×10^{-7} and 7.6×10^{-5} emu/cm^3, with arithmetic mean 1.12×10^{-5} emu/cm^3. The site-mean NRM directions for areas A and B are shown in Figure 2*a*, except for two sites where the samples gave incoherent results. The other 16 site-mean vectors (total 48 samples) lie fairly close to the present earth's field direction. As this could be due to the presence of a secondary component of magnetization capable of masking the primary, a representative set of 38 fresh specimens was demagnetized progressively in alternating fields to 500 Oe (Oe denoting peak oersteds). Typically, up to 150 Oe the NRM directions moved slightly away from the present field direction, remained fixed between 150 and 250 Oe, and became scattered on further demagnetization. A sharp drop of intensity in the 0–200 Oe range indicated removal of a viscous component below 200 Oe. These observations are consistent with the dominance of a stable component between 150 and 250 Oe.

The 112 Mugford specimens previously used for NRM measurement were progressively demagnetized to 100, 150, and 200 Oe. Over this range, the remanence vectors of the specimens from the two sites with incoherent NRM's moved about erratically and J fluctuated widely. This being typical of unstable remanence, the AF results for these sites were rejected along with the NRM data. The best grouping of site mean directions occurred after the 150 Oe step (Fig. 2*b*). The between-site precision improved from $k = 31$ (NRM) to $k = 41$ (150 Oe).

Twelve fresh Mugford specimens were then thermally demagnetized in nitrogen to retard the oxidation of the magnetic mineral. A fairly narrow range of blocking temperatures was observed around 500 °C, indicating a stable component that may correspond to the one isolated after the AF treatment. The thermal treatment caused a displacement of the remanence vectors away from the present pole, similar to that observed in the AF treatment. The between-sample consistency of the data, the good agreement between the AF and thermal results, and the apparent absence of metamorphic alteration of the magnetic material in our samples, all lead us to suppose that this stable component had a primary origin close to the time the lavas erupted.

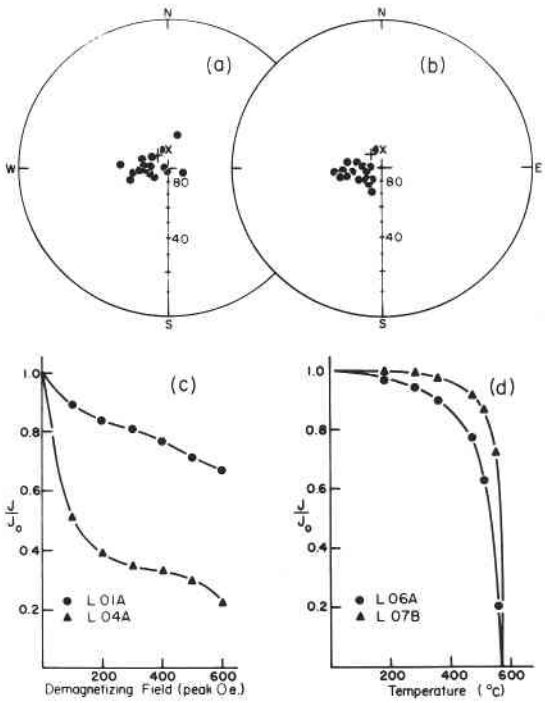


FIG. 2. Some laboratory results. (*a*)(*b*) Site-mean directions of magnetization of the Mugford basalts, excluding two randomly magnetized flows and the two flows of area C. Stereographic (Wulff's) projection, with the present horizontal as circumference. (*a*) Natural remanent magnetization (NRM); (*b*) after alternating-field (AF) demagnetization at 150 peak oersteds. All inclinations are positive (N pole downward). X = Axial dipole field. + = 1968 field. (*c*)(*d*) Normalized intensity, J/J_o, after stepwise demagnetization of two specimens each from Indian Harbour dikes; (*c*) after AF treatment; (*d*) after thermal demagnetization of previously untreated specimens.

Our preliminary microscopic observations of the Mugford basalts indicate that the magnetic material present is magnetite occurring sparsely in polished sections. It is fine grained (typical grain-size is $20 \times 20\ \mu$) and is present as inclusions in pyroxenes. Exsolution or zoning were not observed. In some places magnetite is altered to leucoxene, and chlorite and epidote are also observed. However, there is no evidence from these sections that the bulk of the magnetite, which most probably is the carrier of the observed stable magnetism, is of secondary origin.

The NRM intensities of 16 specimens from the 12 Indian Harbour dikes are distributed between 3.1×10^{-5} and 9.5×10^{-3} emu/cm^3, with arithmetic mean 2.25×10^{-3} emu/cm^3. The NRM directions (not shown here) are quite scattered. AF demagnetization of 10 pilot specimens to 200 Oe apparently has removed a viscous component (Fig. 2*c*). The new directions were better grouped and were hardly affected by further treatment to 600 Oe. These results were qualitatively confirmed by thermal demagnetization in nitrogen of four specimens. High stability is indicated by the thermal decay curves (Fig. 2*d*), which are dominated by a thermally discrete component vanishing close to the Curie point of magnetite. The 16 specimens used in the NRM measurements were AF-demagnetized at 300 Oe, this value being obtained through the optimizing procedure of McElhinny and Gough (1963). The resulting dike-mean remanence vectors mostly pointed southwest with low to intermediate, positive inclinations, but one dike vector was nearly antiparallel to all others. It was inverted in computing the mean direction for the Indian Harbour dike swarm (Table 1).

Table 1 also shows the NRM direction for the 958 m.y. meta-lamprophyre dike. Its deviation from the present field direction and the high precision k both indicate stability. The negative polarity is opposite to that usually reported for rock units of that age. As the direction was obtained only for one dike, no pole position was calculated.

Discussion

The mean remanence direction for the 16 stable Mugford sites (Table 1, first row) is significantly different from both the 1968 field (323°, +79°) and the axial dipole field (0°, +73°). Earlier we stated reasons for supposing that this remanence is primary. The remanence vectors, in order to represent the ambient field direction, must be referred to bedding planes rather than the present horizontal. The mean remanence direction was therefore corrected for the geological dip (16° NW) by making the beds horizontal through rotation about the strike direction (250°). From the corrected direction (Table 1), a paleomagnetic pole relative to the Mugford area was calculated by assuming that the field was a geocentric dipole when averaged over the time of the secular variation. We are satisfied that the sampling of the Mugford basalts

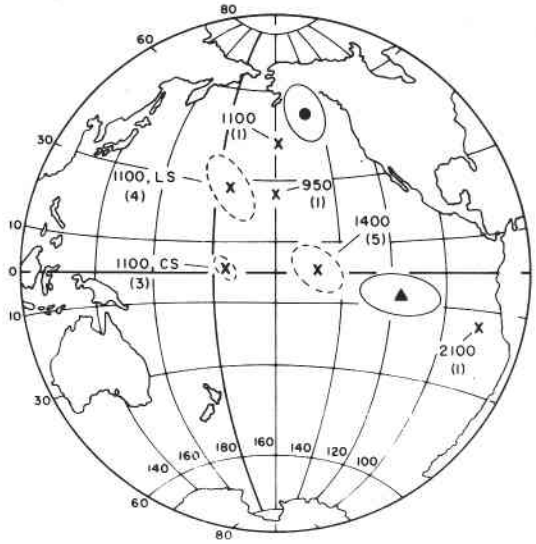


FIG. 3. Pole positions from some North American rocks dated 950 m.y. and older. The poles and 95% confidence ovals for the Mugford basalts (dot) and Indian Harbour dikes (triangle) are compared with mean pole positions (X) obtained from published results for various age groups; ages (950, etc.) are in m.y. Dashed 95% confidence ovals were computed giving unit weight to N separate poles combined in the average (N given in brackets) and rounded to the nearest 100 m.y. References are: *950 m.y.*: Franklin Mountains rock units, Texas (Spall 1971). *1100 m.y., CS* (Canadian Shield): (i) Sudbury dikes (Larochelle 1967); (ii) Mackenzie dikes (Fahrig and Jones 1969); (iii) Muscox Intrusion, and other igneous units (Robertson 1969). *1100 m.y. (1)*: Middle Keweenawan rocks, Lake Superior area; based on two divergent sets of poles of opposite polarity (Palmer 1970). *1100 m.y., LS* (other results, Lake Superior area): (i) Duluth gabbro (DuBois 1962); (ii) Mellen gabbro (Books *et al.* 1966); (iii) Portage Lake lavas (Vincenz 1968); (iv) Minnesota intrusive rocks (Beck 1970). *1400 m.y.*: St. Francois igneous rocks, Missouri (Hays and Scharon 1966); (ii) Igneous rocks, Missouri (Hsu *et al.* 1966); (iii) Front Range igneous rocks, Colorado (Eggler and Larson 1968); (iv) Michikamau anorthosite, Newfoundland (Labrador) (Murthy *et al. 1968*); (v) Croker Island igneous complex, Ontario (Palmer 1969). *2100 m.y.*: Nipissing diabase, Ontario (Symons 1970).

has been extensive enough, vertically and areally, to average out the disturbing effect of the secular variation.

For the Indian Harbour swarm, confidence in the long-term stability of the remanence after AF treatment (Table 1) is justified by the large angle it makes with the earth's field, and by the very stable end points, the improved grouping and the case of a reversed direction obtained through demagnetization. Table 1 also gives a paleomagnetic pole computed from the mean remanence direction.

In Fig. 3, the Mugford and Indian Harbour poles are plotted along with average paleomagnetic pole positions for certain Precambrian ages. We have compiled these from some of the more reliable published data for radiometrically dated rocks in North America. All of the poles appear to be precursors of the North American apparent-polar-wandering curve terminating at the present north pole. Hence the polarity of both the Mugford and Indian Harbour poles is probably 'normal' relative to the present geomagnetic field. Apart from this aspect, a paleomagnetic interpretation of our results will not be attempted here.

Acknowledgments

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Stratigraphy of the Fort Qu'Appelle Vertebrate Fossil Locality, Saskatchewan

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The Fort Qu'Appelle vertebrate fossil locality is in the Echo Lake Gravel, which lies between tills of the Sutherland and Saskatoon Groups. Biostratigraphers have indicated that the Echo Lake Gravel is either a Sangamonian deposit or a Wisconsinan interstadial deposit. The physical evidence however, suggests that the gravel is an outwash deposit which was deposited at the close of an interglacial or interstadial interval during the advance of the glacier that deposited the Floral Formation.

La localité de fossiles vertébrés Fort Qu'Appelle se trouve dans le Gravier Echo Lake, qui est intercalé entre des tills des Groupes Sutherland et Saskatoon. Les biostratigraphes ont indiqués que le Gravier Echo Lake date du Sangamonien ou d'un interstade du Wisconsinen. Cependant, l'évidence physique suggère que le gravier est un dépôt de lavage formé pendant l'avancement du glacier qui a déposé la Formation Floral, à la fin d'un interval interglaciaire ou interstadiale.

Introduction

The Fort Qu'Appelle area (Figs. 1 and 2) has been considered of scientific interest since 1945 when the first vertebrate fossil was found (Khan 1970). Vertebrate fossils have been collected from several gravel pits in the Echo Lake Gravel (Christiansen 1960) extending northwestward along the base of the southwest bank of the Qu'Appelle Valley from the Bliss Gravel Pit to point B in the cross section BB′